

Linux File Permissions & Least Privilege

Hands-on access-control review using ownership, groups, chmod, and validation

Scenario

A simulated shared application directory contains configuration, logs, scripts, and confidential exports. The objective is to grant only required access and validate the resulting permission model.

Path	Required access	Initial issue	Command / action	Validated result
/opt/app/config/app.conf	root read/write; app group read	World-readable configuration	chown root:app app.conf; chmod 640 app.conf	-rw-r----- root app
/opt/app/bin/start.sh	root/app execute; no public write	Group/public permissions too broad	chown root:app start.sh; chmod 750 start.sh	-rwxr-x--- root app
/var/log/app/app.log	service write; security read	Log inaccessible to security group	chown app:security app.log; chmod 640 app.log	-rw-r----- app security
/srv/exports/customers.csv	data owner read/write only	Group-readable sensitive export	chown data:data customers.csv; chmod 600 customers.csv	-rw----- data data
/srv/shared/report.txt	team read/write; others none	Publicly writable	chown analyst:analytics report.txt; chmod 660 report.txt	-rw-rw---- analyst analytics

Validation Checklist

- Use **ls -l** or **stat** to confirm owner, group, and mode.
- Test with representative user accounts when safe to confirm allowed and denied operations.
- Review setuid/setgid bits and world-writable files separately because they can create escalation paths.
- Document why each group exists and who owns membership.
- Re-review permissions after application deployments because package changes can reset ownership or modes.